www.tntextbooks.in

2.8.4 AND operator

The AND operator is defined in Boolean algebra by the use of the dot (.) operator. It is similar to multiplication in ordinary algebra. The AND operator combines two or more input variables so that the output is true only if all the inputs are true. The truth table for a 2-input AND operator is shown as follows:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

The above 2-input AND operation is expressed as: Y = A . B

2.8.5 OR operator

The plus sign is used to indicate the OR operator. The OR operator combines two or more input variables so that the output is true if at least one input is true. The truth table for a 2-input OR operator is shown as follows:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

The above 2-input OR operation is expressed as: Y = A + B

2.8.6 NOT operator

The NOT operator has one input and one output. The input is either true or false, and the output is always the opposite, that is, the NOT operator inverts the input. The truth table for a NOT operator where A is the input variable and Y is the output is shown below:

A	Y
0	1
1	0

The NOT operator is represented algebraically by the Boolean expression: $Y = \overline{A}$

Example:

Consider the Boolean equation:

$D = A + (\overline{B} . C)$

D is equal to 1 (true) if A is 1 or if $(\overline{B} . C)$ is 1, that is, B = 0 and C = 1.

Otherwise D is equal to 0 (false).

The basic logic functions AND, OR, and NOT can also be combined to make other logic operators such as NAND and NOR

2.8.7 NAND operator

The NAND is the combination of NOT and AND. The NAND is generated by inverting the output of an AND operator. The algebraic expression of the NAND function is:

$Y = \overline{A . B}$

The NAND function truth table is shown below:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

A NAND B = NOT (A AND B)